

About Work Study Degree Final Year Project

The Work Study Degree Final Year Project (WSDeg FYP) is an industry-sponsored project provided by the organisation to the student under the programme. The purpose of the WSDeg FYP is to provide the real-world application context to deepen the industry skillsets and apply knowledge to practical business challenges. Through this, it will be mutually beneficial for both NTU and the participating WSDeg industry partners thereby fostering a closer relationship with NTU industry partners.

WSDeg FYP to have NTU students under this programme to work on problems in any of the disciplines through a project requiring application of discipline-related principles & skills, which have a direct industrial significance. The project should take the designing thinking approach of involving an in-depth explorative study, investigation, testing and development to derive a potential solution or product that achieve the business project outcome.

Students are required to submit an interim report, a final report and do a final presentation as part of the academic requirements for the WSDeg FYP. Every WSDeg FYP project will be jointly supervised by both organisation supervisor and NTU school academic staff member.

WSDeg FYP are undertaken by students not on a full-time basis, but with a minimum 1 day per week at the organisation as the students will also be required to study some courses. Therefore, students normally work on the WSDeg FYP in the NTU campus during the semesters. However, periodic visits to companies during the semesters and short periods in companies by students may be arranged, with mutual agreement by industry organisation and its supervisor(s).

The industry organisation is expected to meet the expenses and provide the necessary infrastructure to fulfill the project deliverables. While primary guidance to the student will be provided by the industry mentor with the necessary information & guidance, faculty supervisor will provide the necessary support & guidance for the students to fulfill the academic outcomes.

The project work is essentially that of the student and the industry organisation and the university cannot be held responsible for the accuracy or otherwise of the results.

INDUSTRY-SPONSORED FINAL YEAR PROJECT PROPOSAL FOR WORK STUDY DEGREE PROGRAMME (WSDeg FYP)

Section A : (To be completed by NTU Student)

Name of WSDeg Student : Tan Yi Rong

Matric Number : U2320967K

School of Discipline : School of Mechanical & Aerospace Engineering

Specialisation (if any) Intelligence Manufacturing System

Possible Proposed Faculty Supervisor
(Please leave blank if you wish the School to recommend a suitable WSDeg FYP supervisor)

Section B : (To be completed by Organisation)

Organisation	<u>GlobalFoundries</u>		
Name of Project Proposer & Supervisor & Endorser	<u>Ng, Christopher</u>	Designation	<u>F7 Manager Factory Automation</u>
Email	<u>Christopher.NG@globalfoundries.com</u>	Contact No.	<u>+65 90887159</u>

Section C : Project Details
(To be completed by Organisation)

Project Title Feasibility Study and Prototype Development of a Semi-Automated Preventive Maintenance System for OHT Mark 2 Obstacle Detection Sensors

Project Type (Please select) : ☒ Product-Based ☒ Process-Based ☐ Market Research

Project Focus Area : **Automation, Preventive Maintenance and Industrial Equipment Reliability**

Project Objective for the Organisation	To improve the current preventive maintenance (PM) workflow for Overhead Hoist Transport (OHT) Mark 2 obstacle detection sensors by studying and developing a prototype-level semi-automated PM concept that reduces repetitive manual work, improves consistency of maintenance checks, and supports future integration of docking, sensor verification and maintenance record capture. The project aims to evaluate practical methods of improving the current PM process through suitable mechanical, electrical, and software-assisted solutions, while considering industrial feasibility, operational constraints and long-term scalability. The development will focus on proof-of-concept validation rather than full production deployment.
Project Deliverables/Outcome (ie a business proposal / a prototype/ a specific finding or insight etc.)	The project is expected to deliver a feasibility study and prototype-level semi-automated PM workflow solution for OHT Mark 2 obstacle detection sensor maintenance. The main deliverables will focus on documenting the existing PM workflow, developing an offline software-assisted PM prototype and evaluating suitable connector handling or test mechanism concepts for future implementation. Expected deliverables may include: • A documented study of the current manual PM workflow and its limitations • A proposed semi-automated PM architecture for OHT Mark 2 sensor verification • A comparative evaluation and concept proposal for connector handling or docking methods, which may include robot-assisted docking, guided mechanical fixtures, magnetic docking interfaces or other suitable implementation methods depending on feasibility • An offline software prototype to support PM sequencing, operator prompts, local result logging, screenshot capture and maintenance record generation • A proposed software interface concept for displaying sensor reading information and camera/vision feedback, using simulated or available live sources where feasible • A feasibility evaluation of selected implementation approaches, including hardware simplification, software assistance and PM process redesign • Prototype validation results, limitations and recommendations for future implementation or enhancement The final project outcome is intended to be a proof-of-concept and feasibility evaluation rather than a fully deployed production system. The minimum expected outcome will include a documented workflow study, a software-assisted PM prototype and a feasibility evaluation of suitable connector handling or docking approaches. Additional hardware or integration features, such as live sensor display, camera display, magnetic docking, guided fixture development, or robot-assisted demonstration, may be developed as extended prototype work if time, resources, and technical feasibility allow.

Project Description (Describe a summary of the project scope : What & how the students will be investigating, development/testing and deriving in this project)	This project investigates the feasibility of improving the current preventive maintenance (PM) process for OHT Mark 2 obstacle detection sensors used in the Automated Material Handling System (AMHS) environment. At present, the PM process is carried out manually by technicians, who are required to connect to multiple sensor ports, simulate obstacle and non-obstacle conditions, interpret sensor responses, make adjustments where needed and capture screenshots or records as evidence of the maintenance activity. This manual approach is repetitive, time-consuming and highly dependent on operator handling and judgement. The proposed project will study and develop a semi-automated preventive maintenance concept that improves the consistency and efficiency of this workflow. Rather than defining the solution solely around one hardware platform, the project will investigate multiple feasible implementation methods for reducing manual effort and standardising PM execution. Depending on technical feasibility and industrial constraints, the implementation may involve robot-assisted docking, guided mechanical fixtures, magnetic docking interfaces, switching or breakout circuitry or other suitable mechanisms that simplify connector handling and testing operations. The project will focus on three main areas: 1. <u>PM Workflow Study and Requirement Definition</u> The current PM procedure will be studied in detail to identify repetitive tasks, key limitations, operator pain points, sequence dependencies, and opportunities for automation support. This includes understanding how the obstacle detection sensors are currently connected, how test conditions are simulated, how sensor behaviour is evaluated and how maintenance records are captured. 2. <u>Connector Handling and Test Mechanism Concept Development</u> A practical concept will be developed to reduce the difficulty of repeated connector handling and sensor test setup. This may include the study of a guided docking interface, magnetic connector concepts, or other alternative docking methods that reduce precision burden and improve repeatability. The intent is to evaluate a more robust and practical PM setup that can support future implementation without being restricted to a single automation method. 3. <u>Offline Software Prototype for PM Guidance and Record Capture</u> An offline software prototype will be developed to support the PM process in a restricted enterprise environment without internet dependency. The software may include PM step sequencing, operator prompts, confirmation checkpoints, adjustment and retest flow control, screenshot capture, local run-folder creation, result logging, summary generation and software interface concepts for displaying sensor reading information and camera/vision feedback. The software is intended to function as a technician-in-the-loop PM guidance tool, rather than a fully autonomous control application. This provides a realistic and reliable prototype scope while still offering clear industrial value.
Materials/Equipment Required & Provided	Materials and equipment required / provided may include: • Access to OHT Mark 2 PM workflow information and existing maintenance procedures • Access to obstacle detection sensor interface references and related connector information • Engineering support for studying the current PM process and operational constraints • Access to computer workstations and approved offline software development tools • Mechanical and electrical prototyping support for evaluating docking, connector handling, or test fixture concepts • Access to sensor configuration or monitoring references for workflow simulation purposes

	• Access to suitable fabrication or prototyping resources for concept validation where required • Access to collaborative robot specifications, guided fixture concepts, or vendor technical input where available for feasibility assessment The project is expected to make use of a combination of process study, software prototyping and concept-level hardware evaluation to deliver the proposed outcomes.
Project Proposer Signature :	
Date :	

Please complete form and return to :

WSDP Coordinator, Career & Attachment Office, NTU, Nanyang Avenue, Singapore 639798.

(Email : wmdp@ntu.edu.sg) , at least one month before start of the WSDeg FYP period.